7/05

Inheritance is a mechanism of deriving a new class from the existing class .

Advantages :

1. reusability
2. scratch development time will be save or reduced

Already existing class is called parent/super/base class

newly created class is called child/sub/derievd class

syntax :

mode of inheritance or visibility

```
class Child : MOI Parent
{
      members of child
};
```

What cannot be inherited from parent class?

1. private members
2. constructors and destructor
3. friend functions and friend classes.

protected : protected members of a class are accessible by members of that class,friends and child classes.

What can be inherited to Child classes?
protected and public members

```
class Parent
{     protected : int data1;
      public : void f1(){ };
};

class Child : MOI Parent
{
      int data2;
      public : void f2(){ };
};
```

private

if mod is protected

```
class Child
{  int data2;
   protected :  int data1;
                void f1(){=}
   public : void f2(){=}
};
```

```
class Child
{
  int data2;
  int data1;
  void f1(){}
  public:
  void f2()
     {
};     }
```

if mod is public

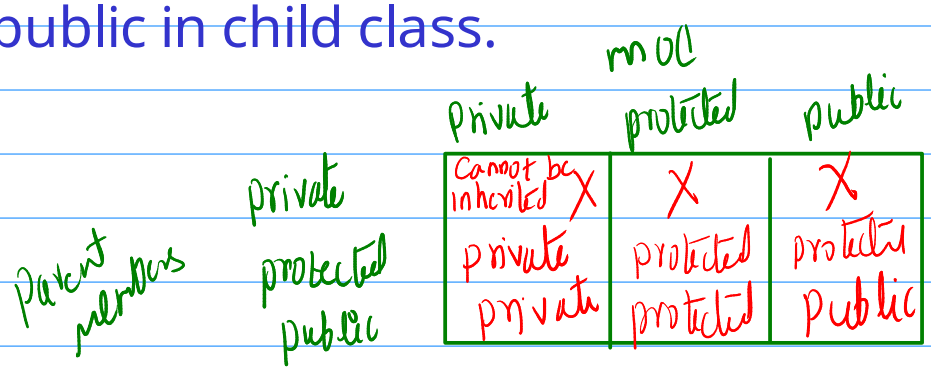
```
class Child
   { int data2;
     protected : int data1;
     public:
        void f1(){ }
        void f2(){ }
};
```

types of MOI
1. private
2. protected
3. public

private mode of inheritance:
if moi is private what ever we are inheriting from parent class becomes private members inside a child class

protected mode of inheritance:
if moi is protected what ever we are inheriting from parent class becomes protected in child class.

public mode of inheritance :
if moi is public , protected members of parent becomes protected and public members becomes public in child class.



size of Child object

sum of the size of parent object,size of individual members of child class

ex :

```
class parent
{ protected : int data1,data2;

};
class child : private parent
{
      int a,b;
};
```

sizeof(child)

sizeof(parent)+sizeof(a)+size of(b);

8 +4+4 = 16

Constructors and Destructors in inheritance

When we create a child object first its parent constructor is invokes then child constructor

When Child object getting destroyed first Child destructor then Parent destructor

Constructors are called in the order of parent to Child and Destructors are called in reverse order ie Child to Parent.